Evaluating State-Space Models for Memory-Efficient 3D Medical Image Segmentation

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Abstract

Medical image segmentation of volumetric data requires models that capture both local spatial structure and long-range contextual dependencies. Transformer-based architectures such as Swin-UNETR address this through windowed self-attention, reducing the quadratic complexity of global attention to linear scaling with respect to input size; however, they still incur substantial memory overhead due to the storage of query, key, and value projections and intermediate attention activations, which becomes significant for high-resolution 3D volumes. In this work, we investigate state-space models as a memory-efficient alternative by performing a controlled architectural modification in which transformer encoder blocks of Swin-UNETR are replaced with Vision Mamba modules, enabling linear-time sequence modeling without explicit attention matrices. Experiments on the BraTS 2021 dataset evaluate segmentation performance using Dice similarity coefficient and Hausdorff distance, alongside computational metrics including GPU memory consumption, parameter count, and FLOPs. Our results show that early-stage integration of Vision Mamba reduces peak training memory by approximately 20%, and peak inference memory by 46.6% confirming its efficiency advantages; however, naive substitution leads to training instability characterized by exploding hidden states and degraded segmentation performance. Further analysis reveals strong stage-dependent trade-offs between memory efficiency and stability, as well as distributional mismatches between transformer and state-space representations, highlighting both the promise and current limitations of state-space models for volumetric medical image segmentation. Our implementation is publicly available at GitHub repository.

1 Introduction

Medical image segmentation is a fundamental task in biomedical imaging, enabling automated delineation of anatomical structures and pathological regions in volumetric scans such as CT and MRI. Early deep learning approaches relied primarily on convolutional neural networks (CNNs), such as 3D U-Net [1], which demonstrated that encoder–decoder architectures with 3D convolutions can effectively capture spatial context in volumetric medical data. However, CNN-based models are inherently limited in their ability to capture long-range dependencies across large spatial regions due to the locality of convolution operations.To address this limitation, transformer-based architectures have been introduced for medical image segmentation. Models such as UNETR [2] and Swin-UNETR [3] incorporate self-attention mechanisms to capture long-range contextual relationships. In particular, Swin-UNETR employs a hierarchical Swin Transformer encoder with windowed self-attention and a U-Net-style decoder, enabling multi-scale feature extraction while modelling long-range dependencies within volumetric data. By restricting attention computation to local windows, Swin Transformer reduces the quadratic complexity of global self-attention to linear scaling with respect to input size. However, despite this improvement, transformer-based models still incur substantial memory overhead due to the storage of query, key, and value projections as well as intermediate attention activations, which becomes significant for high-resolution 3D volumes.

Recently, state-space models have emerged as a promising alternative for modelling long-range dependencies. Mamba [4] introduces a selective state-space formulation that enables linear-time sequence processing while maintaining strong long-range modelling capability. Unlike attention-based models, Mamba processes sequences without constructing explicit attention matrices, significantly reducing memory requirements. This motivates the exploration of state-space models as a more scalable alternative for volumetric medical image segmentation.

In this work, we investigate whether replacing transformer-based components with state-space models can improve computational and memory efficiency for 3D medical image segmentation. We perform a controlled architectural modification in which the transformer encoder blocks of Swin-UNETR are replaced with Vision Mamba modules, while preserving the remainder of the segmentation pipeline. This setup enables a direct comparison between attention-based and state-space-based sequence modelling within the same architectural framework.

2 Related Work

To extend state-space models to visual domains, Vision Mamba [5] adapts selective state-space modelling to image data by converting spatial feature maps into sequences of visual tokens. The Vision Mamba block combines linear projections, depthwise convolution for local feature mixing, nonlinear activation functions, and selective state-space recurrence. This design enables the model to capture both local spatial interactions and long-range dependencies while maintaining linear computational complexity. Several recent works have explored the integration of Mamba-based architectures for medical image segmentation. U-Mamba implements the integration of the Mamba layer into the encoder of nnUNet [6] to improve general medical image segmentation [7], while SegMamba [8] combines the U-shape structure with Mamba for modeling the whole volume global features at various scales for 3D medical image segmentation, and Swin-UMamba [9] introduces a Mamba-based model, designed specifically for medical image segmentation tasks, leveraging the advantages of ImageNet-based pretraining. These approaches demonstrate the feasibility of incorporating state-space modules into segmentation pipelines and highlight their potential for improved efficiency. However, many of these works introduce entirely new architectures or primarily focus on 2D medical images. As a result, a controlled comparison between transformer-based attention mechanisms and state-space models within the same 3D segmentation architecture remains limited.

The primary contribution of this project is to perform a controlled architectural modification in which the transformer encoder blocks of Swin-UNETR are replaced with Vision Mamba blocks while preserving the rest of the segmentation pipeline. In the modified encoder, spatial feature maps will be reshaped into sequences of visual tokens so that selective state-space modelling can be applied. After the state-space computation, the representations will be reshaped back into spatial feature maps, allowing the modified encoder to interface with the existing decoder structure of Swin-UNETR.

3 Method

This section describes the methodological framework, including the preparation process and the organization of the BraTS 2021 dataset, preprocessing procedures, and the training, validation, and testing splits used for internal evaluation. Next, we present the baseline architecture, Swin-UNETR, outlining its hierarchical transformer encoder and U-Net-style decoder design for volumetric feature extraction and segmentation. We then introduce the proposed hybrid model configurations, in which selected transformer encoder stages are replaced with Vision Mamba blocks while preserving the remainder of the segmentation pipeline.

3.1 Dataset

Experiments in this study were conducted using the BraTS 2021 dataset, a benchmark dataset for brain tumor segmentation in multi-modal 3D magnetic resonance imaging (MRI). The publicly available training set contains 1,251 subjects, each consisting of four co-registered MRI modalities: T1-weighted (T1), contrast-enhanced T1-weighted (T1Gd), T2-weighted (T2), and T2 Fluid-Attenuated Inversion Recovery (T2-FLAIR). All volumes were preprocessed by the challenge organizers through rigid registration, skull stripping, and resampling to an isotropic resolution of 1 × 1 × 1 mm, resulting in input volumes of size 240 × 240 × 155. Ground-truth annotations are provided for three nested tumor sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Since the official BraTS 2021 validation and testing benchmarks require submission to an external evaluation server and were not accessible in our experimental setup, we constructed an internal evaluation protocol using the publicly available labeled training data. The 1,251 subjects were randomly partitioned into training, validation, and testing subsets using a 70%/15%/15% split, corresponding to approximately 875, 188, and 188 subjects, respectively. The training subset was used for model optimization, the validation subset for hyperparameter selection and checkpoint selection, and the held-out test subset for final performance evaluation. This internal split enabled a consistent and controlled comparison between baseline and hybrid architectures under identical data conditions.

3.2 Baseline Model

As the baseline architecture, we adopt Swin-UNETR [3], a transformer-based encoder–decoder network designed for volumetric medical image segmentation. To ensure reproducibility and consistency with prior work, our implementation is based on the official MONAI research implementation released by the authors. The overall architecture of the baseline model is illustrated in Fig. 1.

The input to Swin-UNETR is a multi-modal 3D MRI volume $X \in \mathbb{R}^{H\times W\times D\times S}$, where S=4 corresponds to the four MRI modalities in the BraTS 2021 dataset. Following patch partitioning, the input volume is divided into non-overlapping 2×2×2 patches and projected into an embedding space of dimension C=48. The encoder consists of four hierarchical stages, each comprising two Swin Transformer blocks. Within each stage, self-attention is computed over non-overlapping local windows using window-based multi-head self-attention (W-MSA), followed by shifted window multi-head self-attention (SW-MSA) in the subsequent block to facilitate cross-window information exchange. At the end of each stage, a patch merging layer reduces the spatial resolution by a factor of two while increasing the channel dimensionality, enabling hierarchical multi-scale feature extraction. The encoder therefore produces feature representations at progressively lower resolutions across all four stages. The decoder follows a U-shaped design, where encoder feature representations at each resolution are connected to the decoder through skip connections. At each decoding stage, feature maps are first processed using residual blocks consisting of two 3×3×3 convolutional layers with instance normalization, followed by deconvolution-based upsampling by a factor of two. The upsampled features are concatenated with the corresponding encoder features from the previous resolution and further refined using additional residual convolutional blocks. Finally, a 1×1×1 convolutional layer followed by sigmoid activation produces voxel-wise predictions for the three BraTS tumor sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). In our implementation, the baseline model contains approximately 62.19 million parameters, closely matching the original 61.98 million reported [3].

3.3 Hybrid Model Configurations

To investigate whether state-space models can improve memory efficiency in volumetric medical image segmentation, we construct two hybrid variants of the baseline Swin-UNETR architecture by selectively replacing Swin Transformer encoder stages with Vision Mamba (ViM) blocks. The overall hybrid encoder path architecture is illustrated in Fig. 2. In both variants, the patch embedding layer, decoder, skip connections, and all remaining encoder stages are preserved from the baseline architecture, enabling a controlled comparison between attention-based and state-space-based feature modelling. Given an intermediate feature representation $X_i \in \mathbb{R}^{H_i\times W_i\times D_i\times C_i}$ at encoder stage i,

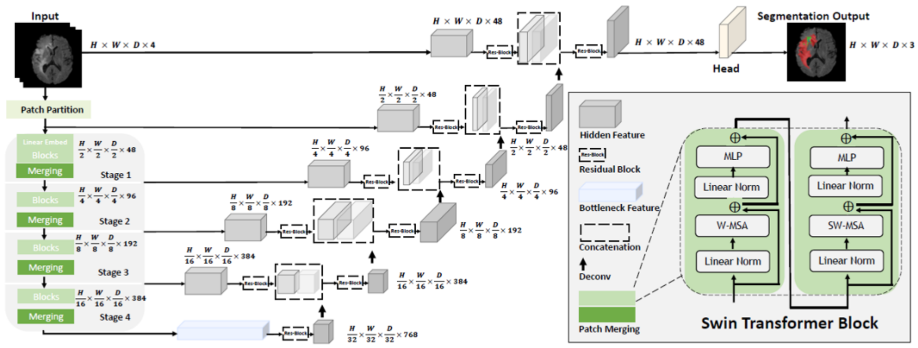


Figure 1: Overview of the baseline Swin-UNETR architecture used in this work. The model consists of a hierarchical Swin Transformer encoder and a U-shaped decoder with skip connections for volumetric medical image segmentation. Reproduced from [3]

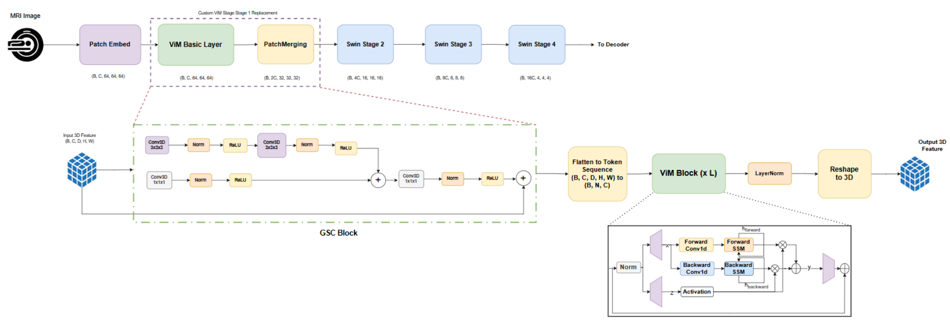


Figure 2: Overview of the proposed Stage 1 hybrid encoder architecture. The first or fourth Swin Transformer stage is replaced with a custom Vision Mamba (ViM) block, while the remaining encoder stages and decoder are preserved from the baseline Swin-UNETR architecture.

the feature map is first processed by a Gated Spatial Convolution (GSC) block to enhance local spatial feature extraction prior to sequence modelling [8]. As shown in Fig. 2, the GSC block consists of parallel 3×3×3 and 1×1×1 convolutional branches with normalization, ReLU activation, and residual connections, allowing local volumetric context to be preserved before tokenization. The resulting feature map is then flattened along the spatial dimensions to form a sequence of $N_i = H_i W_i D_i$ visual tokens, which is passed through the Vision Mamba block.This block performs linear token projections, depthwise convolution for local token mixing, nonlinear activation, and selective state-space recurrence to model long-range dependencies without explicitly constructing attention matrices. Following sequence modelling, the token representations are normalized and reshaped back into their original 3D spatial form before being passed to the subsequent encoder stages.

To analyze the effect of stage placement, two hybrid configurations are evaluated. In the Stage 1 Hybrid, the first Swin Transformer encoder stage operating at the highest spatial resolution is replaced with the proposed GSC–ViM module. In the Stage 4 Hybrid, the fourth encoder stage operating at the lowest spatial resolution is replaced instead. This enables a direct comparison of state-space modelling at both early high-resolution and late low-resolution feature representations.

3.4 Implementation Details

The baseline Swin UNETR and the proposed hybrid models were implemented using PyTorch and MONAI. The learning rate was set to 0.0008. We normalize all input images to have zero mean and unit standard deviation according to non-zero voxels. Random patches of $128 \times 128 \times 128$ were cropped from 3D image volumes during training. We apply a random axis mirror flip with a probability of 0.5 for all three axes. Additionally, we apply data augmentation transforms of random per-channel intensity shift in the range $(-0.1, 0.1)$, and random intensity scaling in the range $(0.9, 1.1)$ to the input image channels, consistent with the original Swin UNETR training pipeline. These preprocessing and augmentation settings were directly inherited from the official MONAI

Table 1: Comparison of baseline validation performance between the original Swin-UNETR implementation and our reproduced implementation on the BraTS 2021 validation split.

Version	Dice Score			Hausdorff Distance (mm)		
	ET	WT	TC	ET	WT	TC
Paper Implementation	0.858	0.926	0.885	6.016	5.831	3.770
Our Implementation	0.855	0.913	0.884	4.745	7.844	5.551

Table 2: Comparison of baseline test performance between the original Swin-UNETR implementation and our reproduced implementation on the BraTS 2021 test split.

Version	Dice Score			Hausdorff Distance (mm)		
	ET	WT	TC	ET	WT	TC
Paper Implementation	0.853	0.927	0.876	16.326	4.739	15.309
Our Implementation	0.866	0.916	0.904	5.215	7.949	5.808

implementation. The batch size per GPU was set to 1. All models were trained for a total of 100 epochs using a linear warmup and cosine annealing learning rate scheduler. Due to computational constraints, training was performed on a single RTXA6000 GPU rather than the 8-GPU DGX-1 setup used in the original implementation. Gradient checkpointing and automatic mixed precision were enabled during training to reduce memory consumption. For validation during training, we use a sliding-window inference approach with an overlap of 0.5 for neighboring voxels. Final post-training evaluation was performed with an overlap of 0.7 for consistency with the original Swin UNETR evaluation protocol.

For the proposed hybrid architecture, the training, preprocessing, and inference settings were kept as identical as feasible to the baseline Swin UNETR to ensure a fair comparison. The only architectural modification was the replacement of the selected Swin Transformer encoder stage with the proposed Vision Mamba-based module incorporating the GSC block.

4 Experiments

To evaluate whether state-space models can serve as a memory-efficient alternative to transformer-based architectures for volumetric medical image segmentation, we conducted a series of controlled experiments comparing the baseline Swin-UNETR model with the proposed Vision Mamba hybrid variants. Experiments were designed to evaluate three aspects: (1) segmentation performance, (2) training stability, and (3) computational efficiency in terms of GPU memory consumption, parameter count, and computational complexity.

4.1 Baseline Validation

Before evaluating the proposed hybrid architectures, we first verified that our implementation of Swin-UNETR closely matched the performance reported in the original work. The baseline model was trained under the implementation settings described in Section 4, using identical preprocessing, augmentation, and optimization procedures. Table 1 and 2 compares our implementation against the original reported Swin-UNETR results on both validation and testing subsets. Our implementation achieved Dice scores of 0.855, 0.913, and 0.884 for ET, WT, and TC respectively on the validation split, which closely matches the reported performance of the original model. Similar consistency was observed on the held-out test split. These results indicate that our training pipeline successfully reproduces the behavior of the original architecture, establishing a reliable baseline for subsequent hybrid comparisons. Representative qualitative segmentation results from the baseline model are shown in Fig. 3.

4.2 Hybrid Encoder Integration

After validating the baseline implementation, we replaced selected transformer encoder stages with the proposed Vision Mamba module. Two configurations were evaluated: Stage 1 Hybrid, where the

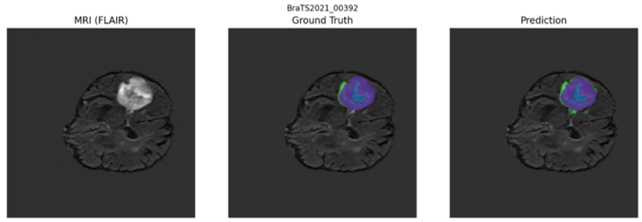
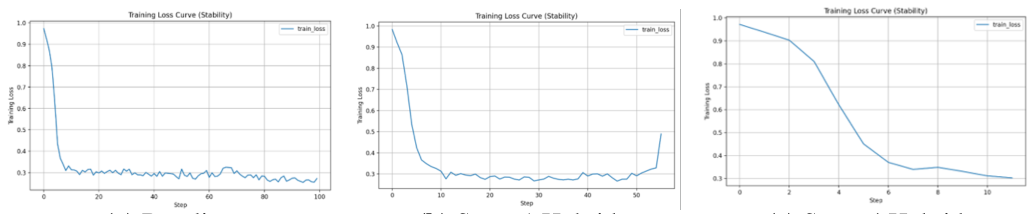


Figure 3: Qualitative baseline segmentation results on a representative BraTS 2021 sample. From left to right: input FLAIR MRI, ground truth, and Swin-UNETR baseline prediction.



(a) Baseline **(b)** Stage 1 Hybrid **(c)** Stage 4 Hybrid

Figure 4: Training dynamics of the three model configurations. (a) Baseline Swin-UNETR shows stable convergence throughout training. (b) Stage 1 hybrid exhibits optimization instability and eventual divergence. (c) Stage 4 hybrid also demonstrates unstable optimization and loss divergence.

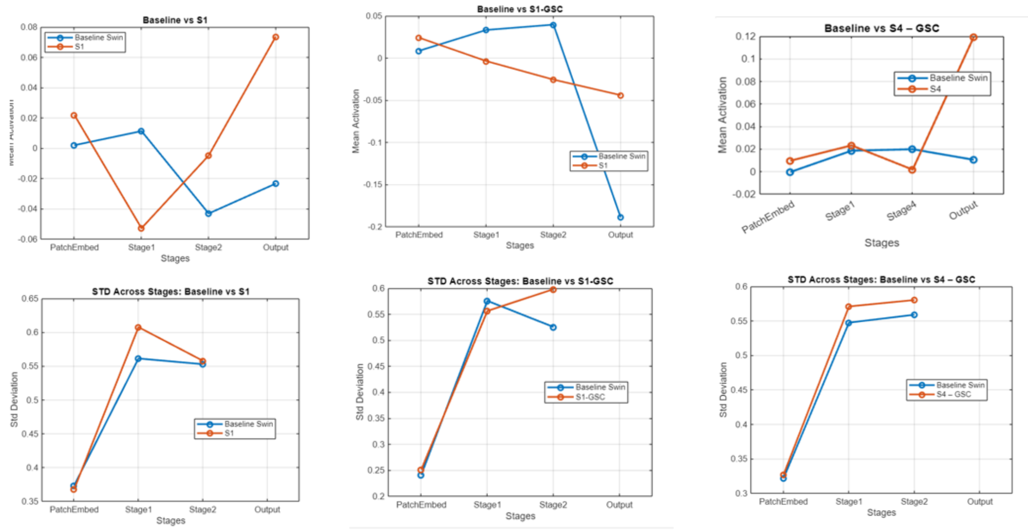
first transformer encoder stage was replaced with the proposed GSC–ViM block, and Stage 4 Hybrid where the fourth transformer encoder stage was replaced with the proposed GSC–ViM block. These two configurations were chosen to study the effect of state-space modeling at both high-resolution and low-resolution feature representations.

4.3 Training Stability Analysis

During training of both hybrid configurations, we observed significant optimization instability compared to the baseline Swin-UNETR model. While the baseline model exhibited smooth convergence across epochs, both hybrid variants frequently encountered exploding hidden states that resulted in NaN values during training. Figure 4 illustrates the training dynamics of all three models. The baseline Swin-UNETR demonstrates stable loss reduction and consistent validation performance, whereas both hybrid variants show sudden divergence during optimization at fairly early stages of training. To identify the source of instability, we conducted ablation studies across hybrid design choices, including: Stage 1 or 4 Vision Mamba integration with and without GSC as listed in Figure 5. Across all configurations, instability persisted, and the ablations show that naive substitution of attention with state-space recurrence introduces representational mismatch with the encoder activations of baseline and hybrid designs. Inspection of intermediate activations revealed abnormal feature distribution growth inside the Vision Mamba hidden states, indicating hidden-state explosion as the primary failure mechanism.

4.4 Segmentation Performance Comparison

Table 3 and 4 summarizes the segmentation performance of the baseline and hybrid architectures. The baseline Swin-UNETR consistently achieved the highest segmentation accuracy across both validation and testing splits. In contrast, both hybrid models showed degraded performance across all tumor regions. The Stage 1 Hybrid achieved Dice scores of 0.827, 0.888, and 0.855 on the validation split for ET, WT, and TC respectively, with corresponding increases in Hausdorff distance. Similarly, the Stage 4 Hybrid produced comparable degradation. These results indicate that although Vision Mamba offers architectural advantages in sequence modeling efficiency, direct replacement of transformer blocks leads to poorer segmentation performance in the current formulation. The increase in Hausdorff distance suggests that boundary localization is particularly affected by the altered feature representations.



(a) Stage 1 **(b)** Stage 1 w/o GSC **(c)** Stage 4 w/o GSC

Figure 5: Layer-wise feature statistics (mean activation and standard deviation) comparing the baseline Swin encoder with hybrid ablations. Irrespective of design choices, the ViM Hybrid model activation statistics diverge significantly from the Swin-UNETR baseline.

Table 3: Segmentation performance comparison of the baseline and hybrid architectures on our BraTS 2021 validation split.

Model	Dice Score			Hausdorff Distance (mm)		
	ET	WT	TC	ET	WT	TC
Baseline	0.855	0.913	0.884	4.745	7.844	5.551
Stage 1 Hybrid	0.827	0.888	0.855	8.802	13.739	10.720
Stage 4 Hybrid	0.828	0.892	0.837	6.034	8.775	7.377

4.5 Training Memory Analysis

To evaluate memory efficiency, peak GPU memory consumption was measured during a full training iteration using identical input patches of size $128 \times 128 \times 128$. Measurements were repeated three times, and the maximum observed allocated and reserved memory values were reported.

4.5.1 Stage 1 Hybrid

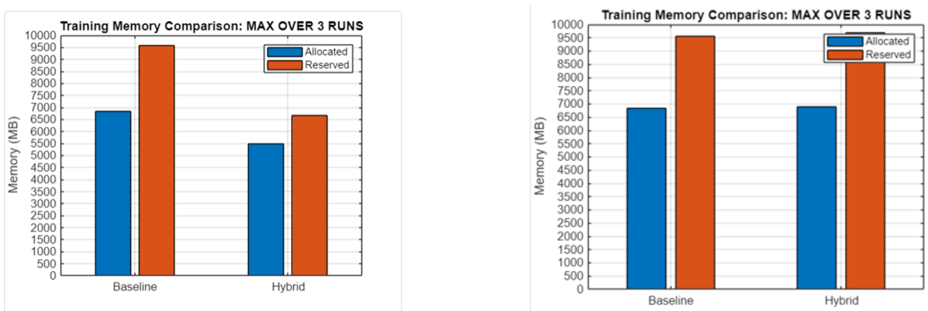
As shown in Fig. 6 (a), replacing the first transformer stage with Vision Mamba reduced peak allocated training memory by approximately 18–20% compared to the baseline model.This reduction can be explained by the fact that Stage 1 operates at the highest spatial resolution, where token count is largest. In the baseline model, attention at this stage requires storing query, key, and value projections, attention score maps, softmax activations and intermediate projection outputs. By replacing attention with state-space recurrence and local convolutional mixing, the hybrid model avoids storing these large attention activations, resulting in substantial memory savings.

4.5.2 Stage 4 Hybrid

In contrast, replacing Stage 4 resulted in a slight memory increase of approximately 1.34% during training as seen in Fig. 6 (b). Layer-wise memory profiling revealed that the additional overhead originated primarily at the hybrid backbone output. When the GSC module was removed, this overhead was significantly reduced, confirming that local convolutional enhancement contributes to the observed increase. Since Stage 4 operates after multiple patch-merging operations, the number of spatial tokens is already heavily reduced, making attention relatively inexpensive at this stage. Consequently, replacing it with Vision Mamba provides limited memory benefit.

Table 4: Segmentation performance comparison of the baseline and hybrid architectures on our BraTS 2021 test split.

Model	Dice Score			Hausdorff Distance (mm)		
	ET	WT	TC	ET	WT	TC
Baseline	0.866	0.916	0.904	5.215	7.949	5.808
Stage 1 Hybrid	0.845	0.887	0.873	9.257	14.279	11.021
Stage 4 Hybrid	0.845	0.890	0.849	7.438	9.731	8.523

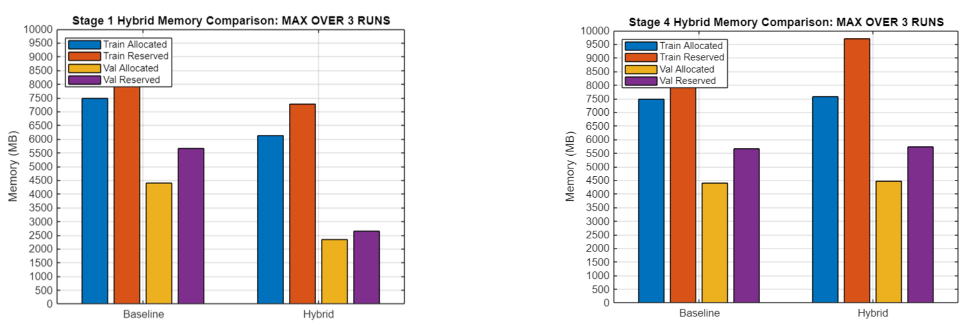


(a) Stage 1 Hybrid **(b)** Stage 4 Hybrid

Figure 6: Training memory comparison (peak GPU memory, maximum over three runs) between the baseline Swin-UNETR and hybrid configurations. (a) Stage 1 hybrid reduces training memory, whereas (b) Stage 4 hybrid produces only marginal memory change.

4.6 Inference Memory Analysis

We next evaluated inference memory consumption using full-volume sliding-window inference on complete BraTS volumes. As shown in Fig. 7(a) Stage 1 Hybrid reduced peak inference memory by approximately 46.6% compared to baseline, representing the largest efficiency gain observed in this study. This suggests that the first transformer stage constitutes a major activation-memory bottleneck during inference, where forward activations dominate memory usage. By contrast, in Fig. 7(b), Stage 4 Hybrid produced a small memory increase of approximately 1.57%, again demonstrating that memory benefits of state-space replacement are highly stage-dependent.



(a) Stage 1 Hybrid **(b)** Stage 4 Hybrid

Figure 7: Inference memory comparison (peak GPU memory, maximum over three runs) between the baseline Swin-UNETR and hybrid configurations using full-volume sliding-window inference on BraTS 2021 volumes.

4.7 Parameter Analysis

Parameter count analysis revealed strong stage-dependent overhead. Replacing Stage 1 increased model parameters by only 0.18%, since the first encoder stage operates at relatively low channel dimensionality. In contrast, replacing Stage 4 increased model parameters by 10.76%. This is because parameter complexity grows with channel dimensionality, and later encoder stages operate at significantly larger feature widths. As shown in Fig. 8, state-space integration becomes increasingly parameter-expensive at deeper stages.

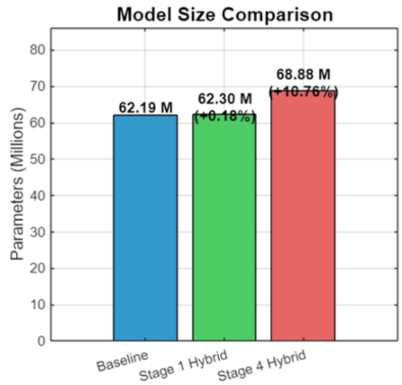
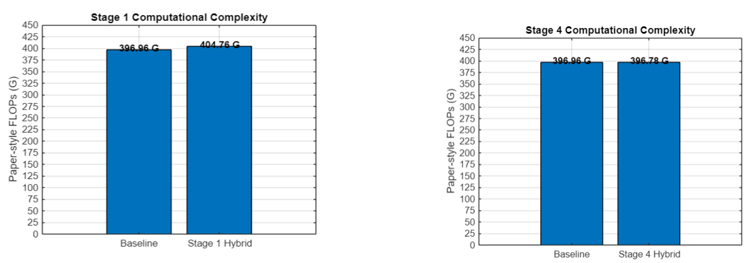


Figure 8: Parameter comparison between the baseline Swin-UNETR and the hybrid configurations.

4.8 Computational Complexity

We evaluated floating-point operations (FLOPs) using a standardized input volume of size 128^3. As shown in Fig. 9(a), the Stage 1 hybrid increased computational complexity by approximately 1.97%, despite achieving substantial memory savings. This suggests that the observed efficiency gains arise primarily from reduced activation storage rather than reduced computation. In other words, the hybrid model trades a modest increase in compute for a significant reduction in memory. In contrast, Fig. 9(b) shows that the Stage 4 hybrid produced negligible computational change, with only a 0.04% reduction relative to baseline. This is expected because both attention and state-space recurrence operate on relatively few tokens at lower spatial resolutions.



(a) Stage 1 **(b) Stage 4**

Figure 9: FLOPs comparison on a 128^3 input volume.

5 Conclusion

Our experiments show that Vision Mamba can provide substantial memory savings when applied to early high-resolution stages. However, naive integration of state-space models introduces significant optimization challenges, including hidden-state explosion, training instability, and distributional mismatch with transformer-based representations. These issues result in degraded segmentation accuracy across all hybrid configurations.

Our findings demonstrate that the benefits of state-space models are highly stage-dependent. Early-stage integration offers meaningful memory advantages, whereas later-stage integration provides limited benefit and may introduce additional parameter overhead. Exploring stabilizing state-space integration through improved normalization strategies, gating mechanisms, and better feature alignment between transformer and state-space representations may enable state-space models to become practical alternatives for memory-efficient large-scale 3D medical image segmentation.

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